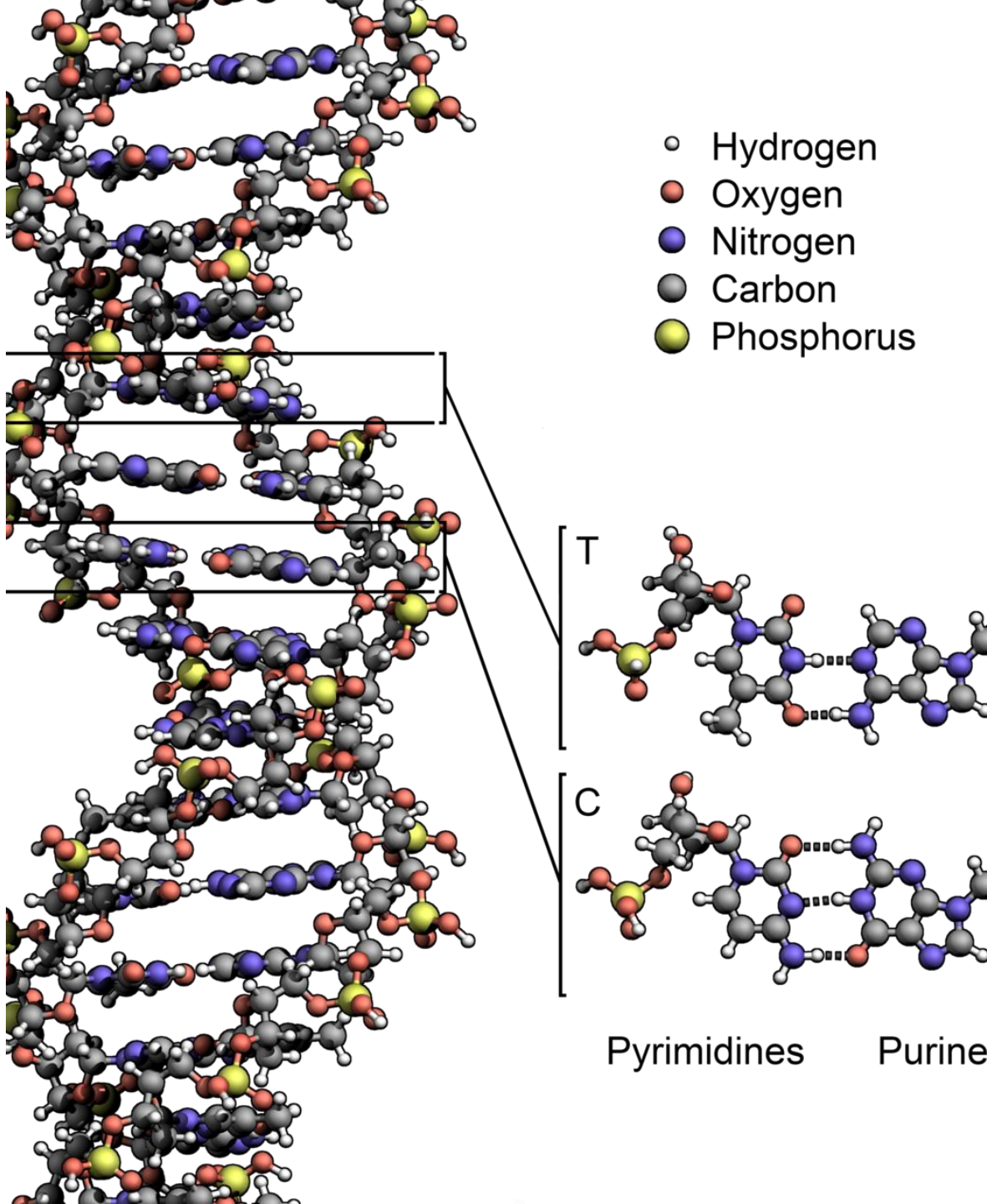
DP Biology – A1.2

Nucleic acids

- How do nucleic acids store information that can be inherited?
- Which features of DNA allow it to be copied accurately?

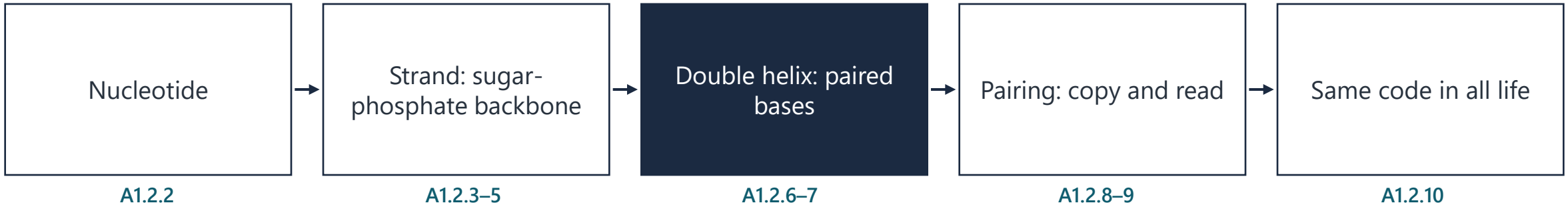
SL statements A1.2.1–10. Textbook pp. 14–24.
Image: structure of the DNA double helix. Zephyris, CC BY-SA 3.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	A1.2.1–5	Draw a nucleotide, and explain how nucleotides join into a strand with a sugar-phosphate backbone
2	A1.2.6–7	Draw a section of DNA, and compare the structures of DNA and RNA
3	A1.2.8–10	Explain base pairing, the capacity of DNA to store information, and the code as evidence of common ancestry

From one nucleotide to inheritance



Nucleic acids are chains of **nucleotides**: a strong, unchanging **backbone** plus a varying sequence of **bases** that carries information.

In DNA, two strands are held by bases that pair only one way, so DNA can be copied and read accurately.

Big question

How can four kinds of building block hold all the instructions to make a human?

DNA is the genetic material of all life

A1.2.1

Every living organism stores its genes as **DNA**.

Some **viruses** use **RNA** instead. They are not usually counted as living: they have no metabolism and reproduce only inside host cells.

Discussion

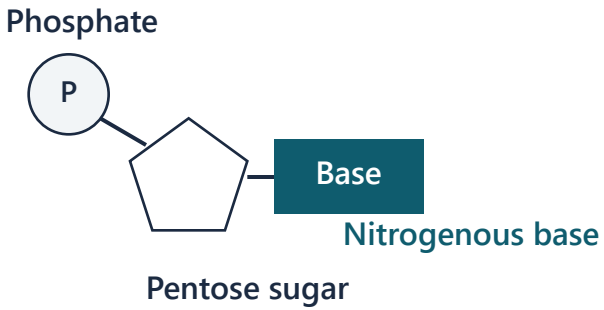
A coronavirus has RNA as its genes. Which characteristics of life does it lack? Does that make it non-living, or a special case?

Where is DNA?

In the nucleus, mitochondria and chloroplasts of eukaryotes; in the cytoplasm of prokaryotes.

A nucleotide has three parts

A1.2.2



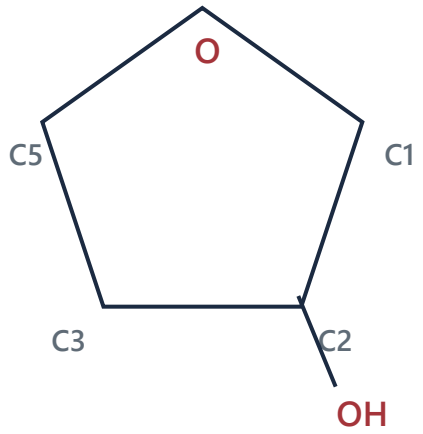
Circle, pentagon and rectangle are enough. Phosphate and base join different corners of the sugar.

Part	What it is
Phosphate	Acidic, negative; links to the next nucleotide
Pentose sugar	Five carbons: deoxyribose (DNA), ribose (RNA)
Base	One of four kinds; carries the information

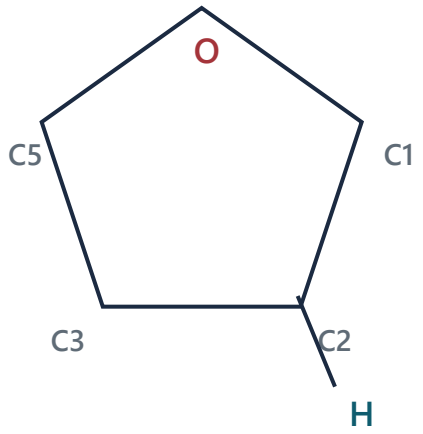
Activity

Draw one nucleotide with the three shapes and label it. Then draw three joined into a chain.

Ribose and deoxyribose



Ribose (RNA)



Deoxyribose (DNA)

Both are **pentoses**: five carbons, four in a ring with an oxygen.

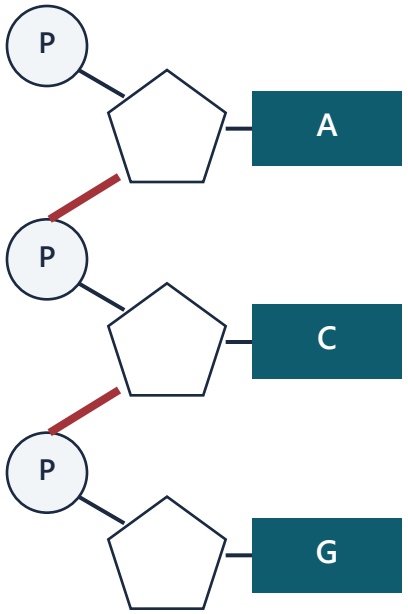
The only difference is at **carbon 2**: ribose has -OH; deoxyribose has -H ("deoxy" = missing an oxygen).

Exam tip

A pentagon with the ring O, and -OH or -H on carbon 2, is enough to show the difference.

The sugar-phosphate backbone

A1.2.3



Red: covalent bonds, sugar to phosphate

The **phosphate** of one nucleotide bonds covalently to the **sugar** of the next, making a **sugar-phosphate backbone**.

The bases stick out to one side. **Any base can follow any other**, so any sequence can be stored.

Key idea

A strong backbone keeps the base sequence intact; the bases carry the information.

The bases: the part that varies

A1.2.4

Base	Rings	DNA	RNA
Adenine (A)	Two	Yes	Yes
Guanine (G)	Two	Yes	Yes
Cytosine (C)	One	Yes	Yes
Thymine (T)	One	Yes	No
Uracil (U)	One	No	Yes

DNA uses A, C, G, T; RNA uses A, C, G, U.

The bases are the only part that varies, so the **base sequence** is the information, like letters in a word.

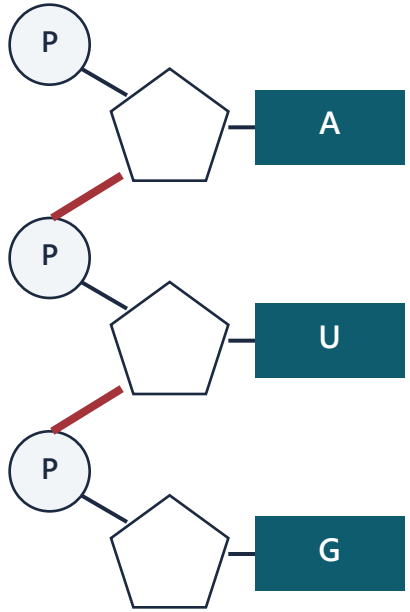
Memory aid

Two-ring purines: A and G. One-ring pyrimidines: C, T, U.

RNA: a polymer built by condensation

RNA is a polymer of nucleotide **monomers**.

Each bond forms by **condensation**: phosphate joins sugar and **water** is released.
RNA is usually a **single strand**.



Drawn RNA: one strand

Try it

How many water molecules are released making an RNA of 100 nucleotides?

Check-in: class 1

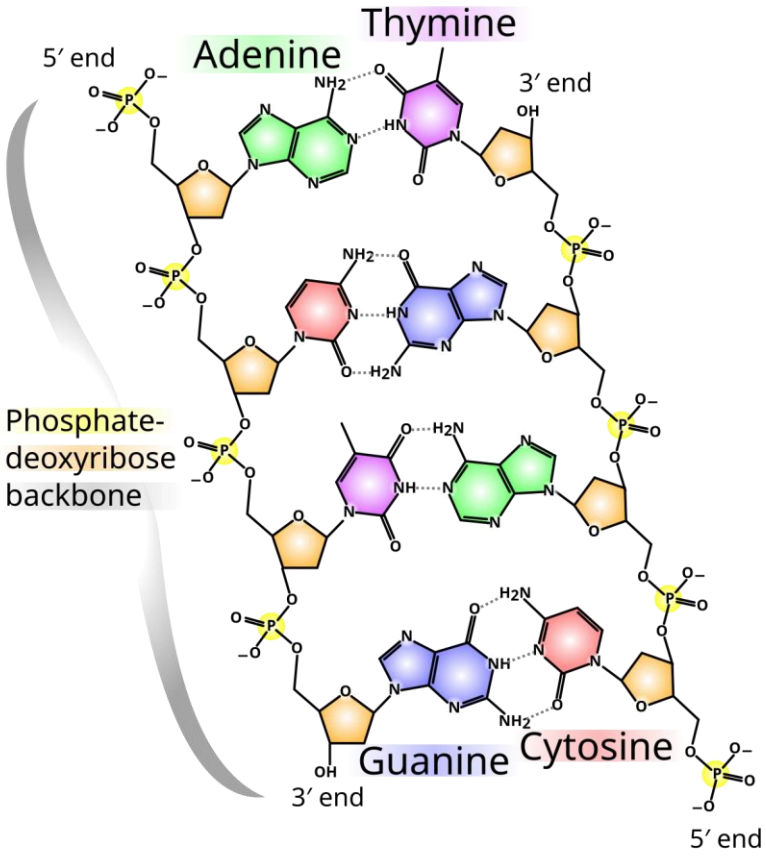
A1.2.1–5

1. Name the three parts of a nucleotide.
2. State the difference between ribose and deoxyribose.
3. Describe how nucleotides join to form a strand.
4. Name the bases found in RNA.
5. Explain why any base sequence can be stored.

How to use this

Answer without your notes, then check against slides 3–9.

DNA is a double helix



Two strands wind into a **double helix**. They are **antiparallel**: they run in opposite directions.

Hydrogen bonds between bases hold the strands together: **A with T**, **G with C**.

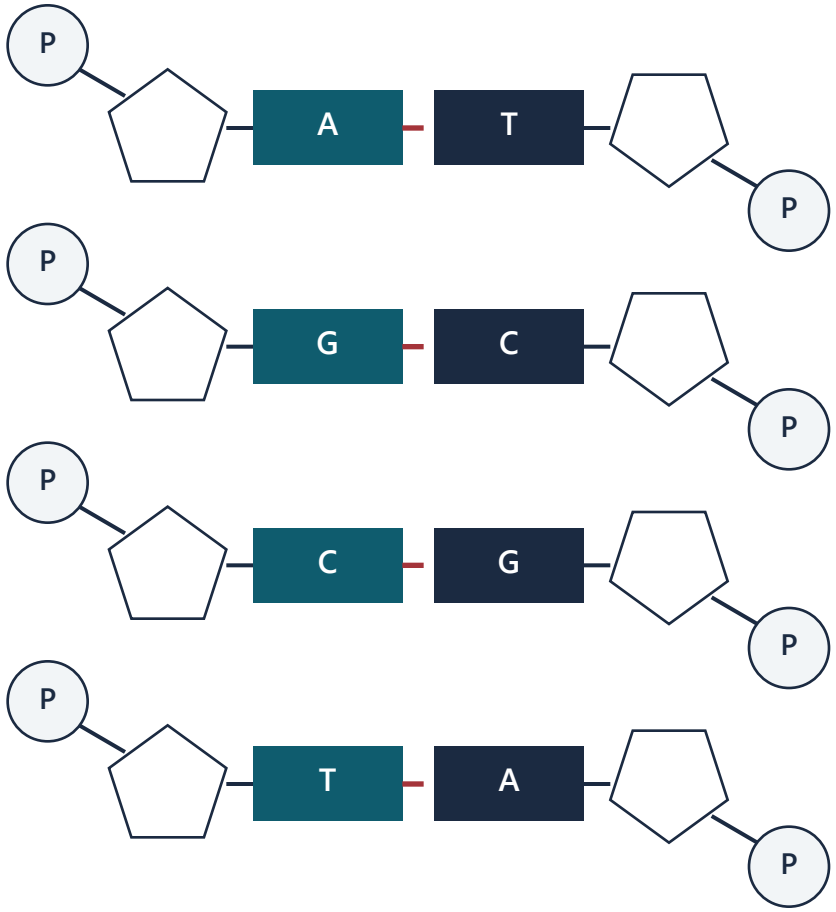
Why this works

Backbones outside protect the bases inside. Each strand holds the information to rebuild the other.

The chemical structure of DNA. Madprime, CC BY-SA 3.0.

How to draw a section of DNA

A1.2.6



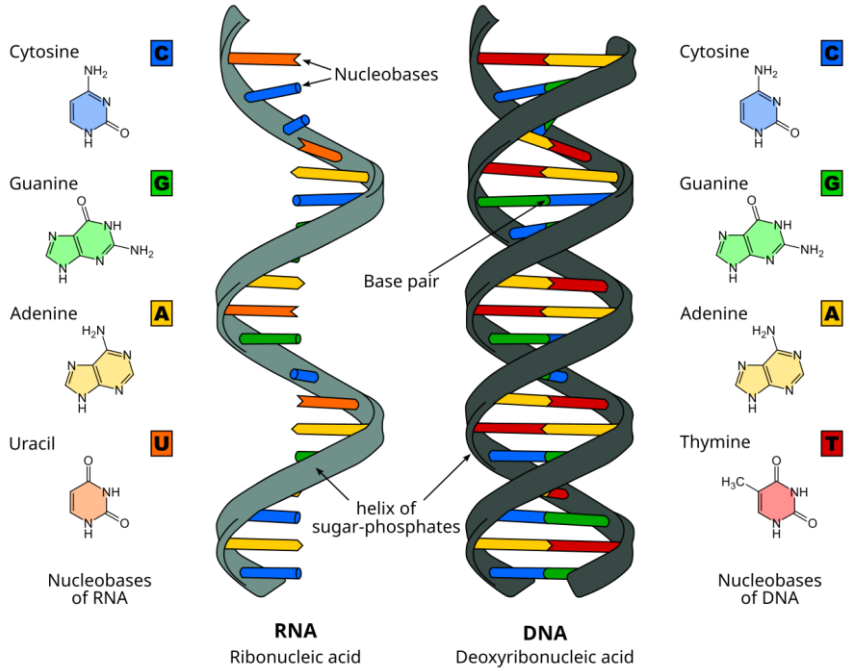
Dashed red: hydrogen bonds

Two backbones; the second strand **upside down** (antiparallel).

Bases paired **A-T** and **G-C**, joined by **hydrogen bonds**; covalent bonds along each backbone.

Comparing DNA and RNA

A1.2.7



Comparing RNA and DNA. Sponk, CC BY-SA 3.0.

Feature	DNA	RNA
Strands	Two	One
Sugar	Deoxyribose	Ribose
Bases	A, C, G, T	A, C, G, U
Role	Long-term store	Working copies of genes

Exam tip

Learn three differences first: strands, sugar, thymine or uracil.

Practice: DNA or RNA?

Sample 1

Contains uracil and ribose; single strand.

Sample 2

A = 30%, T = 30%, G = 20%, C = 20%.

Sample 3

Contains thymine but A does not equal T.

For each one

Decide whether it is DNA or RNA, and whether single or double stranded. Explain.

Working with data: base composition

A1.2.6–8

Percentage of each base in DNA (published values, rounded).

Source	A / %	T / %	G / %	C / %
Human	30.9	29.4	19.9	19.8
Wheat	27.3	27.1	22.7	22.8
Yeast	31.3	32.9	18.7	17.1
E. coli	24.7	23.6	26.0	25.7

1. Compare A with T, and G with C.
2. Explain the pattern using DNA structure.
3. Suggest why values are not exactly equal.
4. DNA is 18% G. Predict % A.

Check-in: class 2

A1.2.6–7

1. Describe the structure of DNA.
2. State what antiparallel means.
3. State the base pairs in DNA.
4. Draw a section of DNA four base pairs long.
5. Give three differences between DNA and RNA.

How to use this

Answer without your notes, then check against slides 11–15.

Why bases pair the way they do

A1.2.8

A two-ring base always pairs with a one-ring base, so every pair is the same width.

Hydrogen bonds form only where the right atoms line up: **A with T** (A with U in RNA), **G with C**. One strand's sequence fixes the other's.

Strand 1	Strand 2
A	T
T	A
G	C
C	G

Try it

A strand reads AATGCCGT. Write the complementary DNA, then the RNA that would pair with it.

Pairing makes copying and reading possible

A1.2.8

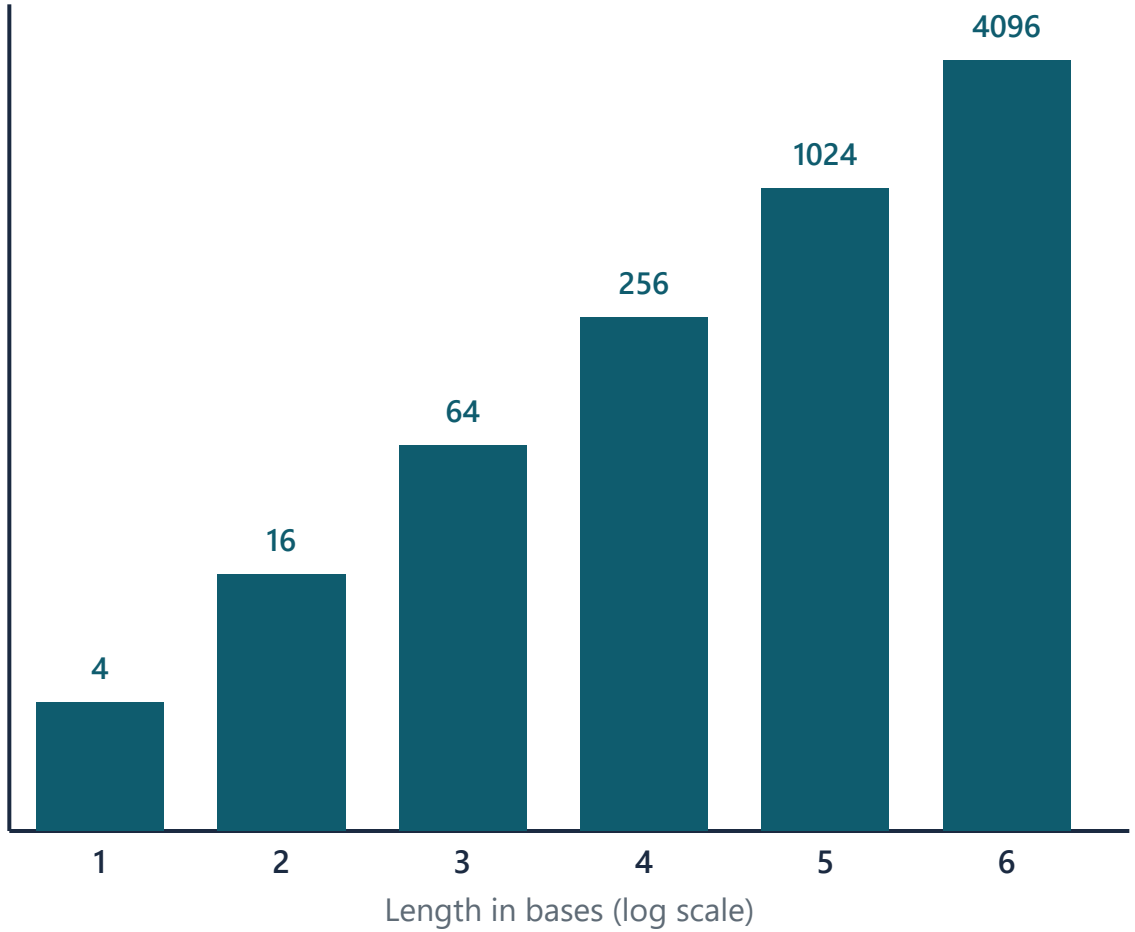
Replication	Transcription	Translation
Each strand is a template; free nucleotides pair with it.	One strand is a template for RNA; U pairs with A.	tRNA anticodons pair with mRNA codons.

Key idea

Complementary pairing is why genetic information is copied with very few errors and read the same way every time (D1.1, D1.2).

An almost limitless store

A1.2.9



Any base can go at each position, so a sequence of **n** bases has **4 to the power n** possible orders.

A human cell has about **3 billion** base pairs per set of chromosomes: the possibilities are unimaginably many.

Big question

Four building blocks are enough because order and length can vary.

Storing data in DNA

Bases	Possible sequences
1	4
3	64
5	?
10	over 1 million

Discussion

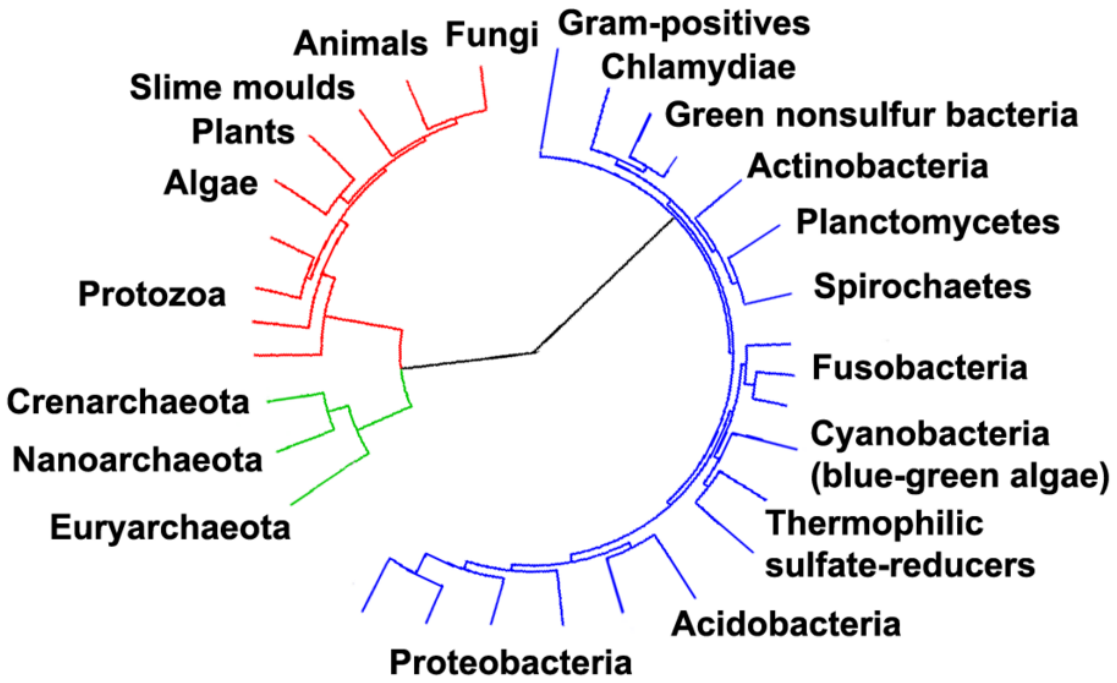
Scientists have stored books and films in DNA. Why might DNA beat a hard drive for storing data for thousands of years?

Try it

Calculate the missing value.

One genetic code for (almost) all life

A1.2.10



A tree of life built from gene sequences. TimVickers, public domain.

Almost every organism links the **same codons** to the same amino acids.

The code is arbitrary, so it is very unlikely to have arisen many times: all life shares a **common ancestor** that used it.

Nature of science

A shared arbitrary feature, like a shared spelling mistake in two essays, points to a shared origin.

Evidence: genes work in other species

A1.2.10



A firefly (Photuris). CC0.

Put the firefly gene for its light-making enzyme into a **tobacco plant**, and the plant glows: it reads the gene with the same code.

Bacteria make **human insulin** from a human gene in the same way.

Link

Gene transfer and protein synthesis: D1.2 and D1.3.

Practice: pairing and the code

A1.2.8–10

Complement

One DNA strand reads GGATCC. Write the other.

Count

How many sequences for a 5-base section?

Explain

Why can a bacterium make a human protein?

Show your reasoning

Use the terms complementary, 4 to the power n and genetic code.

Check-in: class 3

A1.2.8–10

1. State the complementary base pairs in DNA and in RNA.
2. Explain why complementary pairing is needed for replication.
3. Calculate the sequences possible for 4 bases.
4. Explain why the genetic code is evidence for common ancestry.
5. Describe one piece of evidence that genes work in other species.

How to use this

Answer without your notes, then check against slides 17–23.

Common mistakes to avoid

Not this	But this
The bases form the backbone.	The backbone is sugar and phosphate.
A pairs with G, and C with T.	A with T (U in RNA); G with C.
Hydrogen bonds join nucleotides in a strand.	Covalent bonds do; hydrogen bonds join the two strands.
RNA contains thymine.	RNA contains uracil.
The strands run the same way.	They are antiparallel.

Summary: nucleic acids

A1.2.1–10

	Key idea
A1.2.1	All life stores genes as DNA; some viruses use RNA
A1.2.2–4	Nucleotide: phosphate, pentose, base; four bases
A1.2.3, 5	Sugar-phosphate backbone built by condensation
A1.2.6–7	Double helix, antiparallel; DNA v RNA
A1.2.8	A-T, G-C: copying and reading
A1.2.9	4 to the power n sequences
A1.2.10	Shared code: common ancestry

Check that you can...

1. **Draw** a nucleotide and **sketch** ribose and deoxyribose
2. **Describe** how a sugar-phosphate backbone forms
3. **Draw** a section of DNA
4. **Compare** DNA and RNA
5. **Explain** the role of complementary base pairing
6. **Explain** how the genetic code is evidence of common ancestry

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. List the three parts of a nucleotide.

2. State three differences between DNA and RNA.

3. A section of DNA is 4 bases long. Calculate how many sequences are possible.

Image credits

DNA Structure+Key+Labelled.pn NoBB.png – Zephyris – CC BY-SA 3.0

DNA chemical structure.svg – Madprime (talk · contribs) – CC BY-SA 3.0

Difference DNA RNA-EN.svg – File:Difference DNA RNA-DE.svg: Sponk / translation: Sponk – CC BY-SA 3.0

Collapsed tree labels simplified.png – TimVickers at English Wikipedia – Public domain

The Photuris firefly.jpg – Goodymeraj – CC0

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